MODELING AND EVALUATION OF MEDITERRANEAN CORRIDOR: TRANSPORTATION DEMAND, WORKSHOPS, AND DRY PORTS

Abstract:

The development of efficient and sustainable transportation infrastructure is an essential component for economic growth and regional connectivity. In this context, the Mediterranean Corridor project emerges as a strategic initiative to enhance the transportation network in the region, promoting integration and the flow of people and goods. This work focuses on modeling transportation demand, planning railway workshops, and implementing dry ports within the project framework, with the aim of conducting a comprehensive social and private assessment to be used as a pre-feasibility study.

Keywords: Transportation demand modeling, railway workshop planning, dry port implementation, social and private evaluation, pre-feasibility study, Mediterranean Corridor.

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1.- Introduction

The connection of two of Spain's most significant cities, Barcelona, and Valencia, through a high-speed railway network, establishes a strategic bridge linking two of the country's major economic and cultural centers. This connection not only significantly reduces travel times between the two cities but also facilitates the more efficient and sustainable transportation of goods. This has a direct impact on the economy and competitiveness of both the region and the country. The structure of the project is showed in the next figure:



Figure 1. *"Structure of Mediterranean Corridor".*

2.- Transportation Demand Modeling

Transportation demand modeling is crucial for anticipating future needs and optimizing infrastructure planning. Using advanced techniques such as discrete choice models and mathematical optimization, the aim is to understand passenger mobility patterns and cargo transfer along the corridor (Green, P. E., & Srinivasan, V. 1990). This analysis considers factors like population, economic trends, and geographical characteristics to forecast the demand for different modes of transportation, including high-speed trains, buses, and private vehicles. (Agarwal, A., Ziemke, D., & Nagel, K. 2020, Johnson et al 2013, Louviere et al. 2010, Train et al. 2002,2008,2009, Trinca et al. 2001).

2.1.- Utility function

The theory of random utility is based on a proposition from classical economics that assumes individuals assign a certain utility to the characteristics of consumed goods. In this way, random utility posits three principles:

1. Individuals act rationally and possess perfect information, thus choosing the alternative that maximizes their utility given their constraints.
2. If there is a set C of available alternatives, the constraints of each individual n determine the subset $C_n \subseteq$ C of alternatives available to that individual.
3. Everyone perceives a utility U_n for each alternative i (Ortúzar & Román, 2003).

Subsequently, the probability that an individual n chooses alternative i is given by:

$$\mathrm{P}\,(U_{in} > U_{jn}\,) \qquad \forall\, \mathrm{j} \in C_n, \mathrm{j} \neq i \;(1)$$

According to this theory, the utility an alternative provides to a user is governed by the attributes of that alternative. However, it is imprudent to assume that the utility each alternative provides to everyone can be deterministically known, as it may depend on many different factors. In the literature, four possible causes of randomness in utility functions are identified:

1. Attributes that the researcher does not consider in the model, but individuals consider when making their decision.
2. Variation in preferences among individuals that is not considered in the model.
3. Error in the measurements of attributes.
4. The replacement of the measurement of an attribute with some similar variable (Manski, 1977).

Subsequently, to model this randomness, it is assumed that people's utility function can be represented as a deterministic part, observable by the observer, and a random part (McFadden, 1974).

$$U_{in} = V_{in} + \varepsilon_{in} \;(2)$$

And,

$$V_{in} = \sum_{k} \beta_{ik} X_{ikn} (3)$$

This final equation represents observable utility as a function (linear in this case) of the attributes of each alternative, where x corresponds to the value that attribute k takes for individual n in alternative i, and β corresponds to a weight for each attribute and alternative to be estimated.

Note that it is assumed that β takes the same value for everyone, suggesting either homogeneous preferences among the people in the study or successful segmentation based on demographic variables to achieve homogeneity. In the case of transportation, this can generally be achieved using socioeconomic variables (Ortúzar & Román, 2003). Similarly, it is assumed that the relationship is linear, although other specifications can be tested if necessary.

2.2.- Probability of Choosing

Subsequently, the probability of choosing alternative i can be written as:

$$\begin{aligned}(U_{in} > U_{jn}) &= P(V_{in} + \varepsilon_{in} > V_{jn} + \varepsilon_{jn}) \\ &= P(\varepsilon_{jn} - \varepsilon_{in} \\ &< V_{in} - V_{jn}) \\ &= F_{\varepsilon_{jn} - \varepsilon_{in}} \ \forall j \in C_n \ j \neq i \quad (4)\end{aligned}$$

Different models arise based on the assumptions made about the distribution of the error term (ε), as different expressions for calculating the probability of a specific alternative will be derived. One of the most used in estimating modal choice models is the Multinomial Logit (MNL), which arises from assuming that the errors are independently and identically distributed (iid) Gumbel random variables with parameters (0, μ). Thus, the probability of choosing alternative i is given by the expression shown in equation 5, where μ corresponds to a scale factor inversely proportional to the variance of the error term.

$$P(i|C_n) = \frac{e^{\mu V_{in}}}{\sum_{j \in C_n} e^{\mu V_{jn}}} \ (5)$$

Once data on individuals' choices are available, the parameters of the utility function must be estimated. Likelihood, a function that, given a data sample, provides the probability that a set of parameters β generated it, is employed for this purpose. The parameters are then estimated by maximizing this likelihood, resulting in the values that would most likely generate the sample most frequently (Ortúzar & Willumsen, 2011).

Thus, if there are N independent observations on modal choices taken from a choice set C_n, the likelihood is the multiplication of the probability of each choice made, represented by

$$L(\beta) = \prod_{n=1}^{N} \prod_{i \in C_n} P(i|C_n)^{y_{ni}} \ (6)$$

Where Y_{ni} is a binary variable that takes the value 1 if individual n chooses alternative i and 0 otherwise, and P_{ni} is the probability that individual n chooses alternative i. In the case of Multinomial Logit, it would be:

$$L(\beta) = \prod_{n=1}^{N} \prod_{i \in C_n} \frac{e^{\mu V_{in}}}{\sum_{j \in C_n} e^{\mu V_{jn}}}^{y_{ni}} \ (7)$$

2.3.- Subjective Value of Time (SVT)

Subsequently, the SVT is calculated as the marginal rate of substitution between travel time and cost, as shown in the equation 8.

$$SVT = \frac{\frac{\partial V}{\partial t_n}}{\frac{\partial V}{\partial C_n}} = \frac{\beta_{time}}{\beta_{cost}} (8)$$

2.4.- Hypothetical bias

Hypothetical bias can be classified based on its nature, and some types identified by Espino et al. (2004) and Hultkrantz, L. & Savsin, S. (2017). Include:

- Assertion bias.
- Rationalization bias.
- Policy bias.
- No Restriction bias.

Several studies have addressed the existence and impact of this bias. Brownstone and Small (2005) found differences in the value of time between Stated Preference (SP) and Revealed Preference (RP) surveys. Murphy (2005) investigated whether providing information to respondents eliminates hypothetical bias, and while it reduced the bias, it did not eliminate it entirely. Hultkranks and Savsin (2017) conducted experiments with incentives but found issues with bias determination and referencing methodology.

For this study, it will be considered that there is no hypothetical bias when the Subjective Value of Time (SVT) from both revealed and stated preferences are consistent.

2.5.- Pivot

The Pivot method involves presenting the respondent with their chosen alternative in Revealed Preference (RP) and an alternative constructed by modifying the chosen option in RP for each respondent. This creates a Stated Preference (SP) experiment like RP, with two options. One corresponds to the alternative chosen in RP, and the other is a modification of the chosen alternative at a ratio selected by the modeler. Some applications include studies by Rose et al. (2007) and Caussade, Ortúzar, Rizzi, and Hensher (2005).

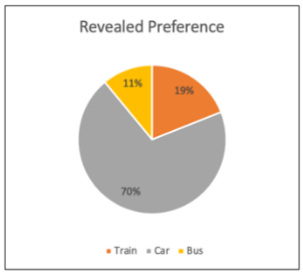
2.6.- Varied Equal Pivot

The elicitation method we will refer to as Equi Varied Pivot is a proposed method where respondents in a Stated Preference (SP) survey are presented with a questionnaire containing two alternatives. One alternative corresponds to the choice in Revealed Preference (RP), and the other alternative is a modification of the chosen RP alternative, but with the same variation ratio for each attribute. In the specific case of the study at hand, the modified RP alternative becomes more expensive and faster in the same proportion. (Flügel, S., Halse, A. H., de Dios Ortúzar, J., & Rizzi, L. I. 2015).

2.7.- K-means

The revealed modal split was extracted from data provided by the Ministry of Transport, Mobility, and Urban Agenda of Spain. However, a critical challenge arises in interpreting origin-destination matrices due to data aggregation in the road mode. The grouping of diverse transportation modes may lead to an oversimplified and inaccurate representation of mobility patterns, distorting the understanding of actual flows and

resulting in suboptimal decisions. To address this complexity, a precise correction is proposed using the mathematical k-means method, allowing for a more accurate allocation of mobility flows to their respective categories by discerning between different modes of transportation, such as buses and cars. This assimilates the periodic behavior of buses with that of trains, considering both as public transportation (Moser et al. 2014). This facilitates a more precise assignment of mobility flows to their respective categories. The revealed modal split is as follows:



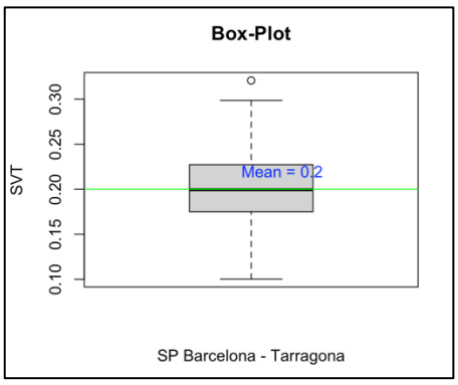
Graphic 1. "*Revealed Preference from Ministry of Transport, Mobility, and Urban Agenda of Spain and k-means disaggregation".*

2.8.- Monte Carlo Simulation

In the field of behavioral research and decision-making, a valuable tool is Monte Carlo simulation applied to logit models. This technique allows for a realistic modeling of how individuals choose among various options based on perceived utility. This utility is derived from key factors such as price and time, which significantly influence decision-making.

The utility function in a logit model encapsulates these influences and is calculated from scale parameters reflecting individual preferences. Monte Carlo simulation enables the exploration of a wide range of possible scenarios by sampling random values for price and time, and then calculating the utility and corresponding choice probabilities for each scenario.

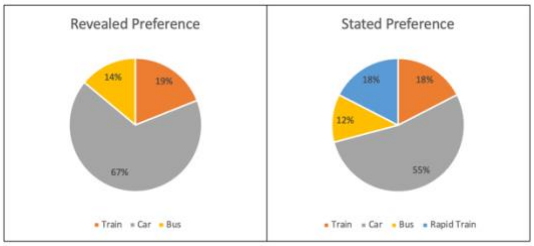
The hypothetical bias is considered nonexistent, as the mean of the estimated SVT is equal to the SVT data simulation generator (0.2 euro/minute), as illustrations the next graphic (Hensher, D. A. 2010).



Graphic 2. *"Hypothetical bias".*

In the Monte Carlo simulation exercise, revealed preference was calibrated by comparing it with the revealed preferences published by the Ministry of Transport, Mobility, and Urban Agenda and disaggregated by K-means.

This calibration process was specifically carried out for the Barcelona - Tarragona stretch, with an average daily annual traffic of 62,844 individuals adjusted to the present from 2017, accounting for a 5% annual demand increase. Additionally, a 30% increase due to demand induction was incorporated, applicable only to the new mode of transportation, resulting in a projection of 103,373 passengers (regardless of the mode). This stretch was chosen exclusively for simplicity and because its traffic flow is evidently superior to other stretches and this flow aligns with the flows carried by a railway system. This selection ensures a more manageable approach and provides a solid foundation for the analysis, focusing on a representative segment that stands out for its relevance in terms of traffic volume and revealed preferences. The results of Revealed and Stated preference are:



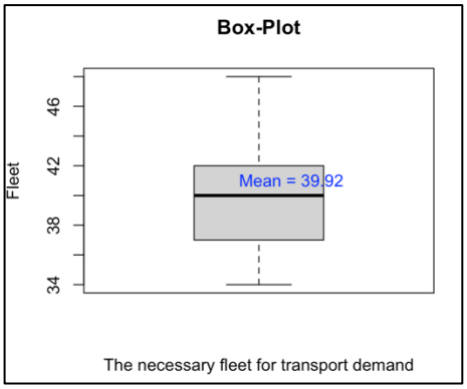
Graphic 3. "*Revealed and Stated preference from Monte Carlo Simulation".*

Both revealed and stated preferences were utilized to construct the social and private project assessment, with the mode shift, as follows in the table 1.

Modal shift		Value
Car	Rapid Train	16%
Bus		19%
Train		15%

Table 1. "*Modal shift from Monte Carlo Simulation".*

The necessary fleet for support this transport demand is 40 trains with a capacity passengers mean of 450. The random variable used to calculate the required fleet is shown in the graph below:



Graphic 4. *"Random variable fleet".*

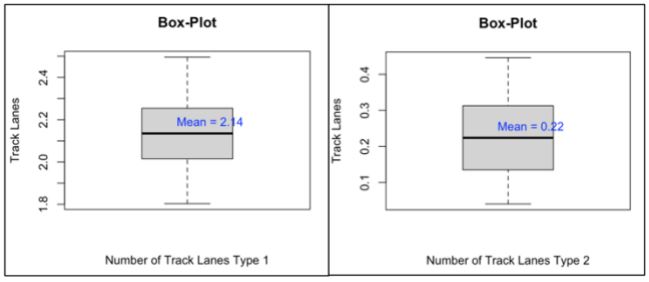
3.- Railway Workshop Planning

Railway workshops play a crucial role in the efficient maintenance and operation of trains. This work addresses the optimization of workshop operations along the corridor. Through resource allocation algorithms and predictive maintenance, the goal is to maximize fleet availability and minimize downtime. Planning includes tactical and operational aspects such as the required number of tracks distribution.

The Poisson process is a statistical tool used to model random events in time or space, breaking down the occurrence of these events into a discrete and independent distribution. In the context of railway workshop design, the Poisson process was applied as a quantitative approach to estimate train arrivals at the workshop and optimize its operation.

The use of the Poisson process in this work focused on modeling the arrival of trains for preventive and corrective maintenance in a railway workshop using an M/M/1 process as described by Larson and Odoni (1983). The arrival rate of trains was established, considering factors such as absenteeism, working hours, and holidays, among others.

The result of the modeling with Poisson processes is shown in the graph below, where the necessary tracks for organizing maintenance workshop tasks around predictive maintenance are observed. It is determined that 3 track lanes are required for preventive maintenance and 1 lane track for corrective maintenance.



Graphic 5. *"Number of track line by maintenance type".*

4.- Implementation of Dry Ports

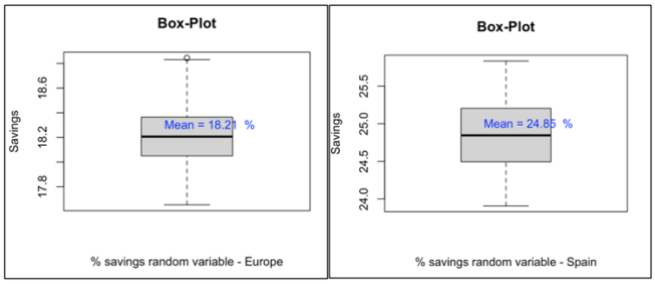
Dry ports, as intermodal nodes, play a crucial role in the efficient transfer of cargo between different modes of transportation, reducing congestion and improving logistics (Muravev et al 2021). This work evaluates the viability of dry ports along the corridor. Considerations include connectivity with the transportation network and benefits in terms of efficiency and sustainability. Implementation of dry ports contributes to optimizing cargo flows and reducing transportation costs.

The linear transportation cost function was optimized to identify the optimal modal split (Bussieck, M. R., Winter, T., & Zimmermann, U. T. 1997), subject to real-world feasibility constraints. This involved fine-tuning the cost parameters to achieve an optimal allocation of transportation modes while ensuring that the modal split remained feasible within the practical constraints of the real-world scenario. The optimization process sought to strike a balance between minimizing costs and adhering to the practical constraints imposed by the existing transportation infrastructure and operational limitations. The result is a refined modal split that represents an optimal distribution of transportation modes considering both economic efficiency and real-world feasibility. The following origin-destination matrix was used for the estimation of cost function savings:"

O/D (MMTON)	Europe	Spain
Barcelona	14	53
Valencia	34	51

Table 2. *"Origin-Destination Matrix of Goods"*

The random variables derived from the optimization of the cost function are shown below. For the purposes of the private assessment, the mean of these random variables will be considered to the private assessment inputs.



Graphic 6. *"Percentage of saving cost – Europe and Spain".*

5.- Social and Private Assessment

Social and private assessment is essential for measuring the economic, social, and environmental impacts of the project (Munda et al. 2004). From a social perspective, a cost-benefit analysis quantifies positive effects in terms of saving time, reduce operation cost, prevention accidents and emission reduction. On the other hand, private assessment focuses on the economic and financial profitability of the project, considering investment, operation, and maintenance costs, as well as expected revenue flows. These complementary assessments allow informed decisions that balance societal and investor interests (Navarro et al. 2006).

6.- Feasibility Study

The work culminates in a comprehensive feasibility study of the Mediterranean Corridor. The collected information, conducted

analyses, and social and private assessments provide a solid foundation to determine the project's viability. This study becomes an essential tool for strategic decision-making, allowing stakeholders to evaluate the economic, social, and environmental benefits of the project. Additionally, the feasibility study contributes to defining the roadmap for more in-depth studies of the Mediterranean Corridor. (Evans, A. W. 2010).

7.- Results

The cost analysis encompassed the Barcelona-Valencia stretch, evaluating various factors, but the derived benefits were specifically attributed to the Barcelona-Tarragona segment. While the cost considerations extended to the entire Barcelona-Valencia route, the focus on benefits pertained exclusively to the Barcelona-Tarragona section. This strategic approach allows for a targeted examination of the costs associated with the broader route while emphasizing the positive outcomes and advantages accruing specifically to the Barcelona-Tarragona connection.

The modeling conducted for the high-speed train and dry port has yielded a holistic financial analysis, encompassing both societal impact and private perspectives of these projects.

In the realm of the social evaluation of the high-speed train, a notable Internal Rate of Return (IRR) of 4% highlights its capacity to generate economic and social benefits exceeding the applied discount rate. Likewise, the calculated Net Present Value (NPV) stands in 2.081 million dollars, reinforcing its positive value both in financial terms and collective well-being.

From a social perspective, the high-speed train project brings significant advantages, such as a substantial reduction in travel times, increasing connectivity, and facilitating people's mobility. Decreasing traffic congestion and reducing harmful emissions contribute to improving air quality and mitigating environmental impact. Furthermore, the generation of direct and indirect jobs during the construction and operation of the train, along with economic stimulation along the route, strengthens social fabric and encourages regional development.

In contrast, the private evaluation of the same high-speed train presents a higher IRR of 19%, indicating greater attractiveness from the perspective of private investors. The corresponding NPV is estimated at 1.470 million dollars (IRR of 9%), confirming the economic viability and return on investment potential for the private sector.

In the context of the dry port, the financial analysis exhibits an IRR of 16%, highlighting its ability to generate attractive financial returns in the long term. The associated NPV amounts to 19.215 million dollars (IRR of 9%), emphasizing its crucial role as an asset and an essential engine for economic and commercial development in the region.

From a social perspective, the dry port entails substantial advantages, such as the decongestion of goods traffic in urban areas, reducing pollution and improving the quality of life for the surrounding community. Additionally, the creation of local jobs, attraction of investments, and optimization of the logistics chain contribute to regional economic growth and overall well-being.

8.- Conclusions

The modeling of transportation demand, planning of railway workshops, and implementation of dry ports are fundamental elements in the evaluation of the Mediterranean Corridor project. The combination of social and private assessments provides a comprehensive view of the project's benefits and challenges, ensuring informed decisions aligned with sustainable development goals.

Collectively, these results highlight the relevance and potential of the high-speed train and dry port projects, both in social and private terms. These quantitative findings reinforce the strategic importance of continuing to explore and drive the implementation of these projects, with the aim of maximizing both social benefits and financial returns.

The favorable results of this feasibility study make it imperative to initiate a formal project study, including citizen participation studies, expropriations, detailed engineering, etc.

Policy must have an informed role in such projects, as strategic decisions must be made, as in this case, to construct the Tarragona – Valencia arc, given the current economic importance of Valencia.

9.- Recommendations

Key recommendations for the successful planning and execution of projects in the context of the Mediterranean Corridor are:

- Comprehensive Collaboration: Foster collaboration between public and private entities to effectively address complex challenges.
- Impact Studies: Conduct comprehensive studies considering long-term economic, social, and environmental aspects.
- Financial Diversification: Explore diverse sources of financing to ensure adequate resources.
- Technological Innovation: Adopt innovative technologies to enhance operational efficiency.

- ❖ Community Involvement: Engage the local community in decision-making.
- ❖ Continuous Monitoring: Establish monitoring and evaluation systems for constant adjustments and improvements.
- ❖ Sustainable Focus: Prioritize environmental and social sustainability at all project stages.
- ❖ Human Resource Development: Invest in the training and development of specialized personnel.
- ❖ Adaptive Flexibility: Design projects that can adapt to future changes.
- ❖ Effective Communication: Clearly communicate social and economic benefits to all stakeholders.
- ❖ Informed Policy: Make strategic decisions in an informed manner.

These recommendations ensure a holistic, sustainable, and efficient approach in the planning and execution of projects in the Mediterranean Corridor region.

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